TEACHER GUIDE

OCR A Level Business (H431)

# Growth and economies of scale

Operations

**Productive efficiency: Economies and diseconomies of scale**



A revision lesson. Pupils run a car dealership in firms of four and revise growth and economies of scale on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

**45-minute setting**

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

### EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

### LESSON SETTINGS

| 30 min | 45 min | 60 min | 75 min |
|---|---|---|---|

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

**Staff · Sabotage** from Round 2



## Finance

**Capital & Loans · Investments**



## Sales & Marketing

**Marketing**



## Operations

**Seasonal stock**

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Unit cost per car

Round 1

A month of sales.

**Unit cost: cost per car sold?**

| | |
|---|---|
| Fixed costs, a month | **£10,800** |
| Variable costs, a month | **£298,260** |
| Cars sold a month | **18 cars** |

**ANSWER**

**£17,170**

(£10,800 + £298,260) ÷ 18 = £17,170

**Why:** Unit cost is all the costs shared between the cars.

## 2 Why unit cost rose

Round 1

After a takeover, your cost per car rose.

**Which two diseconomies are most likely?**

- **A** Fixed costs spread wider
- **B** Staff less motivated
- **C** Bulk-buying discounts
- **D** Cheaper finance
- **E** Poor communication across sites

**ANSWER**

**✓ Staff less motivated**
**✓ Poor communication across sites**

**Why:** Big firms can lose control and morale; the other three cut costs.

## 3 Rank unit cost fall

Round 2

Three plans and what they do to cost per car.

**Order by % fall in cost per car, biggest first?**

- **A** £5,000 down to £4,000
- **B** £2,000 down to £1,500
- **C** £8,000 down to £6,800

**ANSWER**

**1. £2,000 down to £1,500**
**2. £5,000 down to £4,000**
**3. £8,000 down to £6,800**

**Why:** Falls of 25%, 20%, 15%: the biggest fall in pounds is the smallest %.

# Topic tasks

## 4 Unit cost, doubled

Round 2

Selling twice as many.

**Cost per car if sales double?**

| | |
|---|---|
| Fixed costs, a month | **£16,000** |
| Variable cost per car | **£13,550** |
| Cars sold now | **16 cars** |

**ANSWER**

**£14,050**

£16,000 ÷ 32 + £13,550 = £14,050

**Why:** The fixed costs are shared by 32 cars, not 16.

## 5 Stop growing?

Round 3

Each new branch cut your cost per car less, and the last one raised it.

**What should you do next, and why?**

**A** Keep growing

**B** Stop growing for now

**ANSWER**

**✔ Stop growing for now**

**Why:** Rising cost per car is a sign of diseconomies of scale.

## 6 Fixed cost spread %

Round 3

Spread the fixed costs.

**Fall in cost per car, to the nearest %?**

| | |
|---|---|
| Fixed costs, a month | **£34,020** |
| Variable cost per car | **£270** |
| Cars sold now | **14 cars** |
| Cars sold after growth | **18 cars** |

**ANSWER**

**20%**

£2,700 → £2,160; fall 20%

**Why:** The fixed costs are shared by more cars after growth.

# Final questions

Each pupil answers 6 final questions on growth and economies of scale. They come from these 8 questions.

Then each pupil writes one short answer: “Evaluate whether this dealer should keep growing.”

1. A group buys the next town’s family dealership, keeping its staff and site. Which type of growth is this, and its main risk?
   ✔ **External, and the risk is merging two ways of working**

2. A dealer group’s total costs are £2,400,000 for 600 cars sold. After growth they are £3,500,000 for 1,000 cars. By how much does unit cost fall?
   ✔ **£500**

3. A group of ten dealerships orders tyres together and gets a lower price per tyre. Which economy of scale is this?
   ✔ **Purchasing economies**

4. A large dealer group borrows at a lower interest rate than a single-site dealer can. Which economy of scale is this?
   ✔ **Financial economies**

5. After rapid growth, decisions take weeks and branch staff feel ignored by head office. What is this an example of?
   ✔ **Diseconomies of scale from communication and coordination**

6. A dealer opens a second showroom with its own profits and its own staff. Which type of growth is this?
   ✔ **Organic (internal) growth**

7. Two rival dealer groups in the same region combine into one company. What type of growth is this?
   ✔ **Horizontal integration**

8. A dealer group buys the company that supplies its parts. What is the main gain from this backward vertical integration?
   ✔ **More control over supply and its cost**